Interim Design Report

Micromouse X Power Board

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Nyakallo Peete	May 16, 2025
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Chapter 1

Introduction

1.1 Problem Description

A Micromouse is a miniature autonomous robot designed to compete in competitions to solve a maze in the shortest possible time using control-systems. The system is typically divided into four main modules: the motherboard, processor, sensing unit, and power module.

This project focuses on the design and manufacturing of a compact power board which provides reliable and stable power to all components while ensuring efficient operation under specific load conditions.

1.2 Scope and Limitations

1.2.1 Scope

The scope of this project is to design a power board that meets the following eight functional requirements: The scope of this project is to design a power board that satisfies the following eight functional requirements: battery monitoring using an INA219 sensor over the I2C interface; bidirectional control of two main motors (200 mA each) and two auxiliary motors (500 mA each); external switching of two high-side connected loads (1 A each); voltage regulation to provide 5 V (1.5 A) and 3.3 V (300 mA); battery charging via a USB-C host; dual-mode charging at 200 mA and 600 mA; integration of an ON/OFF switch; and support for USB-C communication and power delivery to the rest of the Micromouse. The scope of the project excludes the design and manufacturing of the Micromouse's motherboard, processor, and sensing unit

1.2.2 Limitations

This project is limited to a maximum board size of 82 mm $\times$ 60 mm, and a placement constraint requiring the power board to fit beneath the motherboard without extending to or interfering with its drill holes. Moreover, the power board must adhere to a strict budget cap of $70 throughout the development process. Lastly, all documentation required for manufacturing the board must be completed and submitted over a three-week period and the overall final board must submitted for demonstration by the 14th of May 2025.

1.3 GitHub Link

Access to the GitHub used to collaborate and design the power board may be found via this link: Power Board Project

Chapter 2

Requirements Analysis

2.1 Requirements

The requirements for the Micromouse's power module are described in Table 2.1.

Table 2.1: Requirements of the Power Module

Requirement ID	**Description**
RQ01	The system shall operate up to four motors bidirectionally.
RQ02	The system shall monitor the battery's voltage and current.
RQ03	The system shall support battery charging.
RQ04	The system shall support two battery charging modes: high-current and low-current.
RQ05	The system shall support a USB-C interface.
RQ06	The system shall support external switching of two loads.
RQ07	The system shall provide two regulated voltage outputs.
RQ08	The system shall include a physical ON/OFF switch.
RQ09	The system shall not exceed a manufacturing cost of $70.
RQ10	The system's dimensions shall not extend to the drill holes on the Micromouse motherboard.

2.2 Specifications

The specifications outlined in Table 2.2 are derived from the functional requirements listed in Table 2.1, and define the detailed performance expectations for the Micromouse power module.

Table 2.2: Specifications of the Power Module derived from the requirements in Table 2.1.

Specification ID	**Description**
SP01	Two motors shall draw 200 mA each at the maximum battery voltage of the 1S1P battery.
SP02	Two motors shall draw 500 mA each.
SP03	An INA219 chip shall be used to monitor the battery using via the I2C bus.
SP04	Each external load switching circuit shall supply 5 V at 1 A to the loads.
SP05	The regulated 3.3 V shall supply up to 300 mA, within $\pm$ 5% accuracy.
SP06	TThe regulated 5 V shall supply up to 1.5 A, within $\pm$ 5% accuracy.

2.3 Acceptance Testing Procedures

Table 2.3: Power Module Acceptance Tests

ID	Description	Testing Procedure	Pass/Fail Criteria
AT01	Shunt Voltage Test	Load the INA219 by connecting a dummy load of 100 Ω between the IN- (low-side terminal) pin and ground. Using a DC power supply, apply 3.7 V the IN+ (high-side terminal) pin of the INA219's shunt configuration. The expected shunt voltage ($V_{\text{shunt,exp}}$) is 320 mV [1]. **Measuring the actual shunt voltage (V_{shunt}):** Using test probes, connect a voltmeter in parallel with the shunt resistor configuration using the available test points. Finally, compare the measured shunt voltage with the expected value to verify accuracy.	**Pass:** $V_{\text{shunt}} = V_{\text{shunt,exp}}$, or within 2% of $V_{\text{shunt,exp}}$. **Fail:** Otherwise.
AT02	5 V Regulated Voltage Test	Load the voltage regulator IC by using a breadboard to connect a 3.3 Ω resistor between pin 20 (5 V Out) of the connector and the common GND. Using a DC power supply, apply 3.7 V to the supply pin of the IC using pin 14 (Battery) of the connector as the battery voltage will be boosted to generate the 5 V line. **Measuring the Output Voltage (V_{reg}):** Use test probes to connect a voltmeter in parallel with the load. **Measuring the Output Current (I_{out}):** Use test probes to connect an ammeter in series with the load and GND to measure the current drawn from the voltage regulator.	**Pass:** $V_{\text{reg}} = 5$ V with a 5% deviation. $1.43\,\text{A} \leq I_{\text{out}} \leq 1.58\,\text{A}$ **Fail:** Otherwise.
AT03	Low Current Motor Driver: Output Voltage and Current Test (Loaded)	Connect a 18 Ω dummy load in series with the motor terminals. Supply a 4.2 V on pin 14 (Battery) of the connector. **Measuring Output Voltage:** <u>Motor 1:</u> Supply 4.2 V to pin 28 (Motor1_CTRL1) of the connector and leave pin 30 (Motor1_CTRL2) floating. This is the equivalent of supplying a logic HIGH to pin 28 and a logic LOW to pin 30. Connect a voltmeter in parallel with the load to measure the voltage across the load ($V_{\text{out},1}$). <u>Motor 2:</u> Repeat the steps for Motor 1 to measure the volotage across motor 2 ($V_{\text{out},2}$). **Measuring Output Current:** <u>Motor 2:</u> Measure $I_{\text{out},1}$ through the dummy resistor by connecting an ammeter in series with the resistor and GND. <u>Motor 2:</u> Measure $I_{\text{out},2}$ through the dummy resistor by connecting an ammeter in series with the resistor and GND.	**Pass:** $V_{\text{out},1}$=$V_{\text{out},2}$= +4.2V $0.15\,\text{A} \leq I_{\text{out},1} \leq 1\,\text{A}$ $0.15\,\text{A} \leq I_{\text{out},2} \leq 1\,\text{A}$ (within 2% deviation). **Fail:** Otherwise.
AT04	Motor Driver: Bidirectional Test	Repeat the procedure from AT03. After recording $V_{\text{out,1A}}$ and $V_{\text{out,2A}}$: **Reverse Input Pins:** <u>Motor 1:</u> Supply 4.2 V to pin 30 and leave pin 28 floating. <u>Motor 2:</u> Supply 4.2 V to pin 6 and leave pin 4 floating. **Measure Reversed Output Voltages:** Measure $V_{\text{out,1B}}$ across JP21 and $V_{\text{out,2B}}$ across the loads.	**Pass:** $V_{\text{out,1A}}$ = -$V_{\text{out,1B}}$ and $V_{\text{out,2B}}$ = -$V_{\text{out,2B}}$ (both within 2% deviation). **Fail:** Otherwise.
AT05	External Load Switch Control	**Loading the Switch:** Connect a 4.7 Ω load in series with the output terminal of the switching IC. Apply 5 V to the supply pin as well as a 3.3 V logic control voltage to turn the high side switch IC ON. **Measuring Output Voltage, ON state:** Measure $V_{\text{ON},1}$ across the load using a voltmeter which is connected in parallel. Measure $V_{\text{ON},2}$ across the load using a voltmeter which is connected in parallel. **Measuring Output Voltage, OFF state:** Remove the 3.3 V to resemble a logic LOW signal. Measure $V_{\text{OFF},2}$ across the load using a voltmeter which is connected in parallel. Measure $V_{\text{OFF},2}$ across the load using a voltmeter which is connected in parallel.	**Pass:** <u>V_{CTRL} =3.3 V:</u> 3.3 V=$V_{\text{ON},1}$ =$V_{\text{ON},2}$ <u>V_{CTRL} =0 V:</u> 0 V=$V_{\text{OFF},1}$ =$V_{\text{OFF},2}$ (within 2% deviation). **Fail:** Otherwise.

Chapter 3

Subsystem Design

3.1 Design Decisions

3.1.1 High Side Switching Component

To satisfy requirement RQ06, which specifies the inclusion of two high-side switches on the power board capable of supplying up to 1 A of current (SP04), it was decided that integrated circuits (ICs) would be used to implement these switches. This approach was chosen over constructing discrete high-side switches using passive components, in order to reduce PCB footprint and simplify future troubleshooting and maintenance. Table Table 3.1 presents a comparison between two candidate high side switch ICs: the PBSS5240T and the TPS22917DBV.

Table 3.1: Comparison of PBSS5240T and TPS22917DBV ICs.

Feature	**PBSS5240T** [2]	**TPS22917DBV** [3]
Switching Element	PNP BJT	P-Channel MOSFET
Operating Voltage	Up to 60 V	0.8 V to 5.5 V
Maximum output voltage	Up to 60 V	6 V
Output Current	$I_{C(max)} = 6$ A	2 A
Control Logic	Must be designed separately	Included internally
Board Space	Requires more board space for supporting components (i.e. npn transistor and other passive components)	Requires less board space due to its compact design
Protective Circuitry	Must be implemented externally	Included internally
Simplicity	Must design control circuitry separately.	Control logic is included internally
Power Dissipation	Higher: $V_{CE(sat)}$ losses	Lower: low RON MOSFETs

Both ICs meet the current specification of being able to supply 1 A and they both offer high side switching. However, the TPS22917DBV was chosen for its low on-resistance n(RON), higher efficiency, and simplified control compared to discrete BJT-based options like the PBSS5240T. The fact that the TPS22917DBV has a low RON is beneficial as this makes it have the highest chance of passing ATP05. Additionally, it does not require continuous base current and includes integrated control logic, reducing the need for external components and saving PCB space. Its built-in protection features (e.g., slew rate control, reverse current blocking, and output discharge) further simplify the design and improve system reliability.

3.1.2 Motor Controlling Component

To satisfy RQ01, which specifies bidirectional control of four motors with up to 500 mA for auxiliary motors and 200 mA for the others, it was decided that two dual-channel, bidirectional motor driver ICs (H-bridges) would be utilised. Each IC will be assigned to a motor pair, ensuring that the design can pass the relevant tests outlined in ATP03 and ATP04. Table Table 3.2 presents a comparison between two candidate motor driver ICs: the DRV8833PWPR and the TB6612FNG.

Table 3.2: Comparison of DRV8833PWPR and TB6612FNG ICs.

Feature	DRV8833PWPR [4]	TB6612FNG [5]
Types of Motors Supported	Brushed DC and Bipolar Stepper	Brushed DC and Bipolar Stepper
Number of output channels	2	2
Operating Voltage (VM)	2.7 V to 10.8 V	2.5 V to 13.5 V
Operating Current (per channel)	1.5 A	1.2 A
Peak Operating Current	2 A	2 A
Package	HTSSOP-16 (has a thermal pad)	TSSOP-16 (does not have a thermal pad)

Both IC's rated output current per channel are able to meet specifications SP01 and SP002. However, the DRV8833PWPR IC was selected over the TB6612FNG due to its higher output current per channel capacity of 1.5 A per channel. Its HTSSOP-16 package with a thermal pad also offers superior heat dissipation, improving thermal performance, efficiency, and reliability for motor control.

3.1.3 Voltage regulating Components

To satisfy specifications SP05 and SP06 under requirement RQ07, which requires regulated 5 V (1.5 A) and 3.3 V (300 mA), two voltage regulation approaches were evaluated: a single buck-boost regulator and a combination of separate buck and boost converters. Table Table 3.2 presents a comparison between two candidate motor driver ICs: the TPS63020DSJR (buck-boost IC) and the discrete buck & boost IC system.

Table 3.3: Subsystem acceptance tests

Feature	TPS63020DSJR [6]	Separate Buck & Boost ICs
Total Number of IC Types needed	1	2
Total Number of ICs Needed	2	2
Output Voltage Range	-0.3 V to 7 V	Each buck and boost converter's output voltage is dependent on the minimum headroom required between the input voltage and required voltage.
Maximum Output Current	2 A	2 A for each buck/boost ICs
Cost (BOM)	Lower: Extension fee paid once.	Higher: Extension fee paid twice.
Integration	Simplest setup	Complex setup
Power Efficiency	Higher: seamless transition between bucking / boosting.	Lower: Higher chances for noise and heat to be introduced.
Reliability	Higher: Depends on feedback resistor configuration and can manage a smaller headroom	Lower: The 3.3 V regulated voltage would need to be attained by bucking a an already boosted voltage (regulated 5 V) as it cannot handle a small headroom.

The TPS63020 offers efficient and seamless regulation due to its ability to automatically switch between buck and boost modes. This ensures that there is a stable output across a wide input range. Thus, having high chances of passing

ATP02's pass criteria. Using a single IC type for both voltage regulations reduces the complexity of the sub-module and minimizes BOM costs. It performs reliably in low headroom conditions which is beneficial for regulating 3.3 V from the battery supply of between 3.7 V and 4.2 V. Additionally, its high efficiency and fewer components lead to lower thermal buildup and reduced noise, improving system reliability. Using the same IC for both voltage regulation stages simplifies testing and troubleshooting.

3.2 Final Design

3.2.1 Final Design Description Summary

To satisfy RQ01, RQ02, RQ06, and RQ07, several components were selected based on their alignment with key specifications. For RQ01, two dual-channel `DRV8833PWPR` ICs were chosen due to their ability to supply up to 1.5 A, meeting SP01 and SP02. To address RQ02, the `INA219AIDCNR` was selected for current sensing over I2C as specified in SP03 and because its compact SOT-23 package also conserves PCB space. To meet RQ07, two `TPS63020DSJR` buck-boost converter ICs were used. This IC will either buck the battery voltage to 3.3V so that SP05, SP06, and RQ09 are satisfied. For RQ06, two `TPS22917DBV` high-side switches were chosen to meet SP04 and the criteria outlined in subsection 3.1.1.

Lastly, each IC was configured using the recommended typical applications example supplied in its datasheet.

3.2.2 Final Schematic and PCB Images

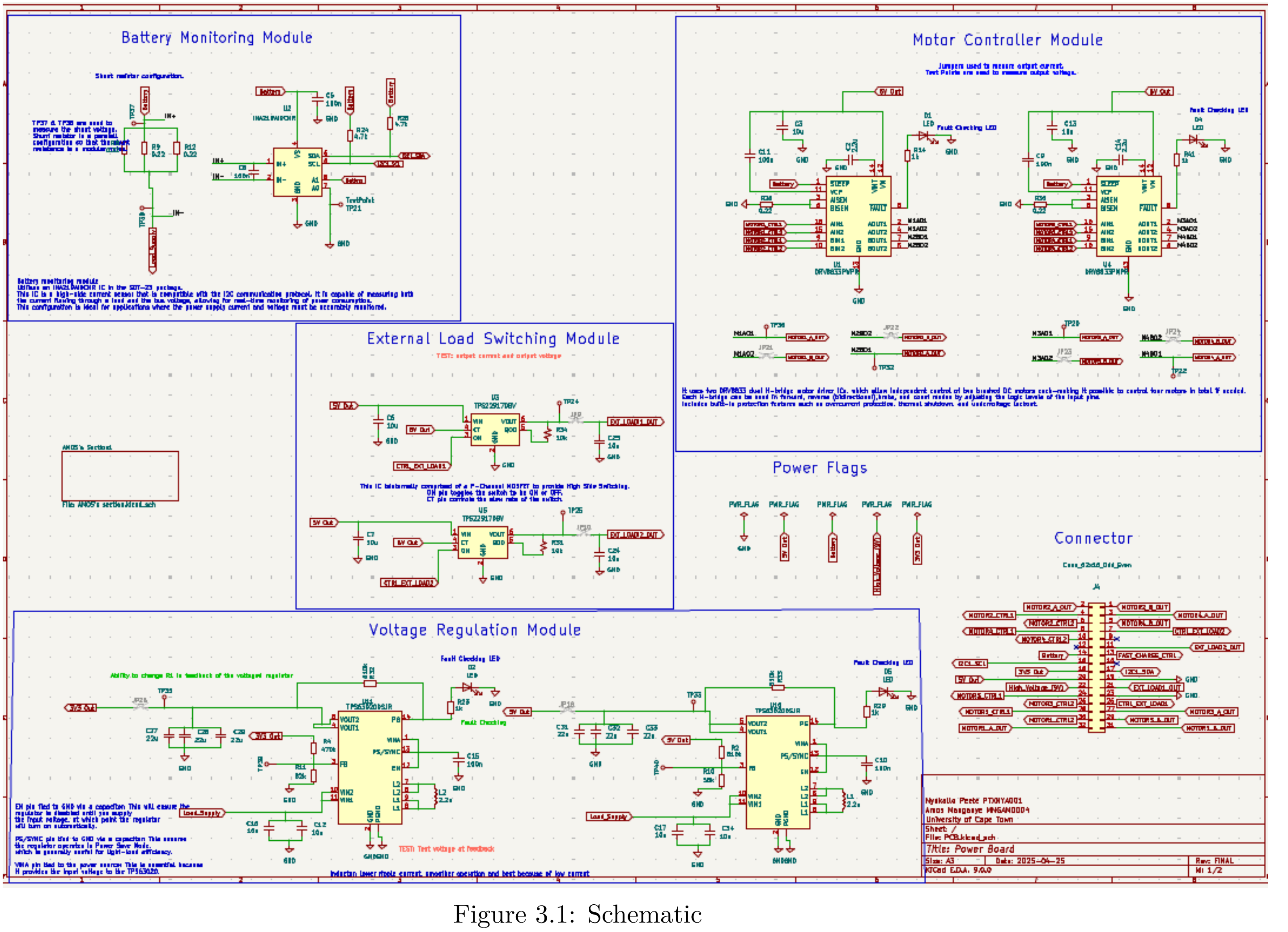
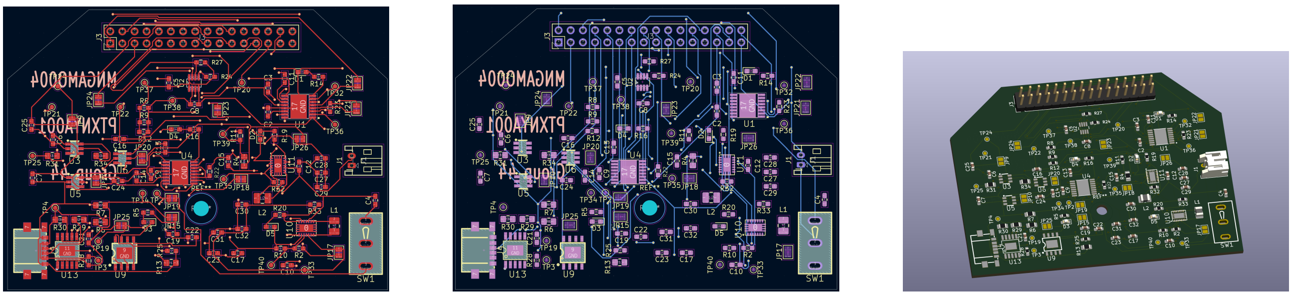


Figure 3.1: Schematic



(a) Front PCB (b) Back PCB (c) 3D PCB

Figure 3.2: PCB

3.3 Failure Management

Table 3.4 describes three failure management features that were implemented on the power board.

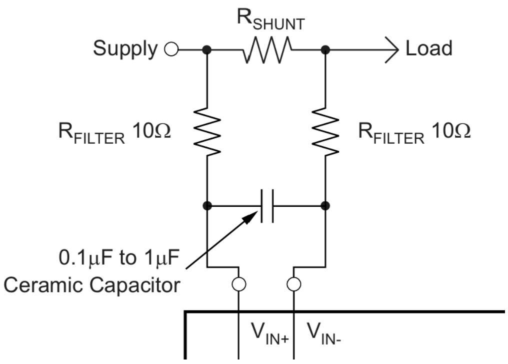


Figure 3.3: Recommended Shunt Noise Protecion [1]

Table 3.4: Power Board Failure Management Features

Name	Description	Purpose
Fault Detection LEDs	An LED with a 1 kΩ series resistor is tied to each IC's open drain status pin (DRV8833PWPR FAULT and TPS63020DSJR PG). These pins are pulled high (logic 1) when the IC is operating normally and pulled low (logic 0) on fault, so the LEDs are illuminated in the healthy state and dont illuminate when in a fault state.	Provides an immediate visual fault indication for the motor driver and buck boost regulator, greatly simplifying troubleshooting.
INA219 Dynamic Shunt Resistance	As discussed in subsection 3.2.1, the shunt resistor for the INA219 was calculated to be 0.1 Ω. To implement fault management, three 0.22 Ω resistors were placed in parallel, resulting in an effective shunt resistance of 0.073 Ω.	For accurate measurements, the shunt resistance must be large enough to ensure accurate measurements by the INA219AIDCNR, but small enough to minimize power loss. The design starts with a smaller shunt resistor, and if it is found to be too small for accurate measurements, a trace can be scratched to leave two resistors in parallel, which results in an effective shunt resistance of 0.11 Ω. This method provides flexibility in adjusting the shunt resistance for optimal performance. A three-pin jumper and a two-pin jumper were intended to be used for this configuration, however, due to the lack of board space, the cutting of traces had to be settled for. The initial design was to utilise a three-pin and two-pin jumper configuration to provide selectable resistance paths, however, due to space constraints on the PCB, this led to a compromise where trace cutting was used instead.
INA219 Dynamic filtering network	To improve current measurement accuracy, a noise filtration network was designed at the shunt input of the INA219AIDCNR, using 0 Ω through-hole resistors (2010 package) as placeholders for the 10 Ω filter resistors (R_{Filter}). This allows dynamic configuration based on noise performance. The chosen resistor type ensures easy soldering and desoldering for adjustments. Due to PCB space limitations, the full filtration network couldn't be implemented. Figure 3.3 shows the recommended shunt configuration, where R_{Filter} would be 0 Ω if implemented.	The goal of this filtration network was to mitigate high-frequency noise that may affect shunt voltage readings. While a shunt capacitor is typically sufficient for filtering, the addition of series resistors allows further tuning of the filter response, especially in electrically noisy environments.

3.4 Bill of Materials

Below is a table that clearly outlines the bill of materials used to make the motor controlling, battery monitoring, voltage regulating and external load switching circuits of the power board.

Table 3.5: Bill of Materials.

Component	**Type**	**Quantity**	**Unit Price [$]**	**Extension Fee [$]**	**Total Cost [$]**
Capacitors	- 10 uF -100 nF -2.2 uF -22 uF	-20 -12 -4 -12	-0.0060 -0.0024 -0.0062 -0.0086	0.00	0.0464
Diodes	Red	8	0.0056	0.00	0.0448
Inductors	2.2 uH	8	0.0402	3.00	3.3216
Resistors	-510 kΩ -470 kΩ -56 kΩ -82 kΩ -0.22 Ω -1 kΩ -4.7 kΩ -10 kΩ	-6 -2 -2 -2 -20 -8 -4 -4	-0.0066 -0.0036 -0.0011 -0.0010 -0.0038 -0.0010 -0.0011 -0.0017	-0.00 -0.00 -0.00 -0.00 -3.00 -0.00 -0.00 -0.00	3.1462
DRV8833PWPR	IC	4	0.6900	3.00	5.7600
INA219AIDCNR	IC	2	0.5845	3.00	4.1490
TPS22917DBV	IC	4	0.2393	3.00	3.9572
TPS63020DSJR	IC	4	0.5910	3.00	5.3640
					25.7892

Out of a total bill of $59.99 to manufacture 2 boards, it cost $25.79 to purchase the materials for which meet the following requirements: RQ01, RQ02, RQ06 and RQ07.

Given that each partner had a budget of $28 to build the four requirements they were responsible for, the final cost price was well within budget. However, a few components had a minimum quantity order which has been taken into account. For example only 4 inductors were needed for the production of two boards, yet a total of 8 inductors were purchased. And, with respect to the 0.22 Ω sense resistors, only 10 were required for the project yet a total of 20 were purchased.

3.5 Micromouse System Interfacing

3.5.1 Interfacing Diagram

Figure 3.4 below describes how the Power Board interfaces with the entire Micromouse system.

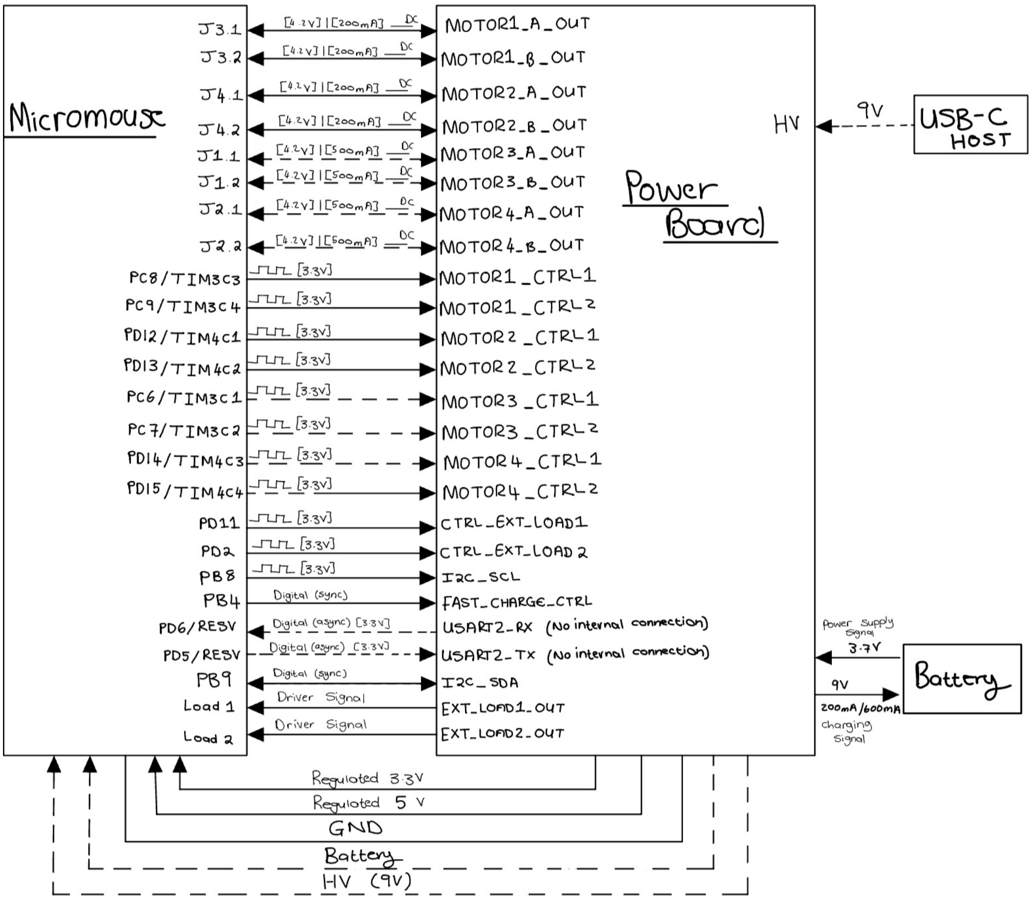


Figure 3.4: System Interfacing Diagram

3.5.2 Interfacing Table

Table Table 3.6 below shows the pin connections between the Power Board and Motherboard (MB). Pins labeled with the abbreviation ‘MB’ adjacent to them indicate that that pin is on the Motherboard, indicating that the pin it is connected to resides on the Power Board.

Table 3.6: System Interfacing Table

ID	Description	Connection between Motherboard and Power Board
IO01	Motor Driver output signals to Motors for bidirectional control	• MOTOR1_A_OUT → J3.1 (MB) • MOTOR1_B_OUT → J3.2 (MB) • MOTOR2_A_OUT → J4.1 (MB) • MOTOR2_B_OUT → J4.2 (MB) • MOTOR3_A_OUT → J1.1 (MB) • MOTOR3_B_OUT → J1.2 (MB) • MOTOR4_A_OUT → J2.1 (MB) • MOTOR4_B_OUT → J2.2 (MB)
IO02	Control signals to Motors Drivers for bidirectional control	• PC8/TIM3C3 (MB) → MOTOR1_CTRL1 • PC9/TIM3C4 (MB) → MOTOR1_CTRL2 • PD12/TIM4C1 (MB) → MOTOR2_CTRL1 • PD13/TIM4C2 (MB) → MOTOR2_CTRL2 • PC6/TIM3C1 (MB) → MOTOR3_CTRL1 • PC7/TIM3C2 (MB) → MOTOR4_CTRL2 • PD14/TIM4C3 (MB) → MOTOR4_CTRL1 • PD15/TIM4C4 (MB) → MOTOR4_CTRL2
IO03	External Load Control and Output Signal	• EXT_LOAD1_OUT → (MB) • CTRL_EXT_LOAD1 → PD11 (MB) • EXT_LOAD2_OUT → (MB) • CTRL_EXT_LOAD2 → PD2 (MB)
IO04	I2C Communication Line	• I2C_SDA ↔ PB9 (MB) • PB8 (MB) → I2C_SCL
IO05	**Reserved** USART2 connection for asynchronous data transmission	• USART2_RX → PD6/RESV (MB) • PD5/RESV (MB) → USART2_TX
IO06	Control Signal for Fast charge control of the battery	• PB4 (MB) → FAST_CHARGE_CTRL
IO07	Power signals and common GND between the Motherboard and Power board	• GND — (MB) • HV_9V — (MB) • 5V — (MB) • 3.3V — (MB)

Chapter 4

Results and Discussion

4.1 Shunt Voltage Test

According to RQ02, the system must have the ability to monitor the battery's voltage and current.SP03 specifies this requirement must be fulfilled an INA219 IC which uses the I2C bus. AT01 outlines the testing procedure used to verify that the theoretically expected shunt voltage ($V_{\text{shunt,exp}}$) closely matches the actual measured shunt voltage (V_{shunt}). This verification is crucial, as it also confirms whether the chosen shunt resistor allows the INA219AIDCNR to accurately satisfy RQ02.

To perform the test, the testing procedure for AT01 was followed closely. The measured voltage across the INA219's shunt configuration was 30 mV.

The initially expected value of 320mV is approximately 966.67% higher than the measured value of 30mV. This discrepancy is attributed to the fact that the datasheet for the IC was misread. A shunt voltage of 320mV belongs to the INA219B variant whereas the selected component was the INA219AIDCNR (INA219A), whose shunt voltage is 32 mV.

Despite the initial misinterpretation, AT01 is considered a pass. The measured shunt voltage deviates by 3% from the expected 32 mV, which is within the 4% tolerance defined by the pass criteria.

4.2 5V Regulator: Voltage and current test

AT02 outlines the testing procedure used to verify that the TPS63020DSJR can boost the typical battery voltage on 3.7 V and supply a regulated 5 V output at 1.5 A with a deviation within 5%.

In order for the test to be conducted, JP18, which was intended to be used to measure the output current needed to be soldered shut as its placement resulted in the feedback configuration of the IC to be compromised. Thereafter, the test procedure described in AT02 was meticulously followed. During the test, the output voltage was measured to be 5.098 V, and the output current was 1.46 A. According to the defined pass criteria, the voltage regulator successfully met performance expectations. This confirms that SP06, which is derived from requirement RQ07, is satisfied.

4.3 Low Current Motor Diver Loaded Test

AT03 outlines the testing procedure used to verify that the motor driver (DRV8833PWPR), which controls the non-auxiliary motors, successfully outputs a voltage closely matching the highest voltage of a 1S1P battery. It also ensures that the motor is capable of drawing 200 mA, as specified in SP01. This specification stems from requirement RQ01.

An error was made during the design phase whereby the motor driver's supply pins (VM and VCP) were connected the regulated 5 V line. To rectify this error, the trace connecting the pins to the incorrect voltage was cut and wires were soldered between the pins and the battery voltage line to correctly interface them. After following the testing procedure for AT03, the following results were recorded:

An output voltage of 4.18 V was measured at a current of 198 mA. These values fall within the acceptable range defined by the pass criteria, thereby confirming that the device meets the requirements.

4.4 Low Current Motor Diver Bidirectional Test

AT04 is used to verify that the motor drivers (DRV8833PWPR) can control the brushed motors bidirectionally. This ensures that the motor drivers fulfill the intended motion functionality of the Micromouse. The functionality is defined in RQ01 which specifies that the motors must support bidirectional control.

The test was conducted following the testing procedure for AT04. The voltage measured when the input supply pins are reversed was -4.18 V.

This result confirms that the motor driver is capable of driving both motor channels in opposite directions, due to the presence of a voltage which is equal in magnitude but opposite in polarity. Thus negative voltage thus validating the system's ability to control each motor bidirectionally.

4.5 External Load Switch: state test

The purpose of this test was to verify the switching functionality of the TPS22917DBV ICs, as required by RQ06. A voltmeter was connected to the corresponding external load output pin on connector J3 to confirm that the IC correctly toggles its output in response to a control signal applied to the ON pin (pin 3).

The CT pin was connected to the VIN pin via a resistor in error. This was rectified by removing the CT pin from the IC to sever its connection as this pin was meant to be connected to GND.

When a supply voltage of 5.01 V was supplied to the VIN pin a logic control voltage of 3.3 V was applied to the ON pin, an output voltage of 5.00 V was measured at the switch's output indicating that the switch was enabled and conducting as expected. When the 3.3 V logic control signal was removed from the ON pin (left floating), the output voltage dropped to 0 V, confirming that the switch was disabled and no conduction occurred.

The results of the test confirm that the TPS22917DBV functions as intended by enabling the output only when the ON pin is driven high (3.3 V supplied). The measured output of 5.00 V when the switch was enabled indicates minimal voltage drop, which is due to the ICs low on-resistance. When the control signal was removed, the output voltage dropped to 0 V, verifying that the switch fully disables the output as expected. These results confirm compliance with RQ06. No unexpected behavior was observed, and the switch response time appeared immediate under test conditions.

Chapter 5

Conclusion

The purpose of this project is to design, implement, and test a robust power module for the Micromouse system. The power module is essential for delivering stable and reliable power to all electronic components to enable the Micromouse to autonomously navigate a maze. The motivation behind this design is rooted in the need for efficient power distribution, the need for accurate battery monitoring, and dependable motor control.

The final solution is an affordable, compact and reliable power board that successfully meets the specifications and requirements outlined in chapter 2. It integrates many key functionalities such as: accurate battery monitoring via the INA219; provides bidirectional control of four motors using two dual-channel DRV8833PWPR ICs; external high-side switching for peripheral loads using P-MOSFET-based switching ICs; and delivers regulated 5 V and 3.3 V rails using the TPS63020DSJR IC.

As discussed in chapter 4, all sub-modules passed their respective acceptance tests, confirming that the design functions as intended. Minor design modifications were made during testing, however, these did not affect overall functionality.

In conclusion, the power module fulfils the project description as its cost price of $59.99 which is well below the $70 budget, it's shape and size is compact and fits beneath the Micromouse's motherboard, and reliably supplies power to all other modules of the Micromouse: the motherboard, processor, and sensing unit.

5.1 Recommendations

For future iterations of the design, several improvements are recommended:

- **PCB Layout Optimisation:** Group interdependent sub-modules closer together to simplify post-production rework and signal routing.
- **Sub-Module Isolation:** Incorporate jumpers between sub-modules to allow for isolated testing and fault diagnosis.
- **Connector Labeling:** Clearly label all connector pins on the PCB silkscreen to improve assembly efficiency and reduce wiring errors during integration and testing.

These changes would enhance both the usability and maintainability of the power module in future development cycles.

Chapter 6

Reflection

6.1 Improvements Across Design Phases

Table 6.1: Improvements

Pase	Description
Design	Begin by thoroughly understanding all design requirements to avoid wasting time during component selection. Predefine testing procedures early in the process to guide design decisions. Ensure familiarity with all required design software before starting to avoid delays later. Lastly, progress must be shared with all partners involved in the project to ensure that all important information is shared across the team.
Board Layout	Minimise the use of jumpers unless they are necessary for testing or failure management. Strategically group related components and route traces efficiently to enhance testability and reduce board size. Label each connector pin so that they are easily identifyable during the testing stage.
Testing	A detailed documentation of all tests and modifications must be maintained. This will ensure that issues can be traced and resolved efficiently, and supports accurate debugging and future iteration.
Report	The report must be consistently updated throughout the project rather than at the end. This ensures that critical details are not forgotten and improves the quality and completeness of the final submission.

6.2 Post-Facto Gantt Chart

The Gantt chart in Figure 6.1 illustrates the actual time allocation and task interdependencies during the course of the project. It provides insight into project pacing and highlights opportunities for better time management in future iterations.

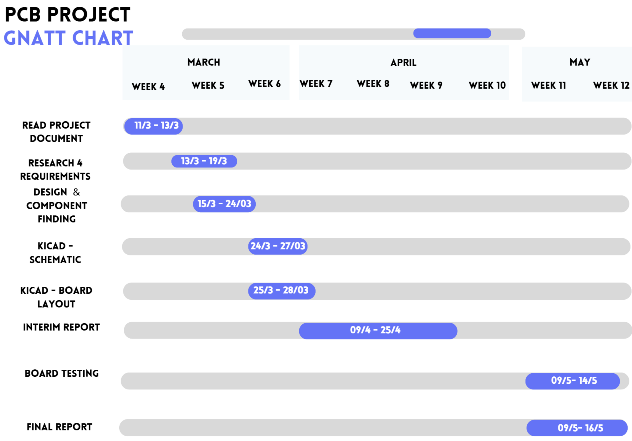


Figure 6.1: Post-facto Gantt Chart

6.3 Cost Optimisation Strategies

To reduce the overall cost of the design, it is important to avoid components with minimum order quantities that exceed the required usage, as this leads to unnecessary payments on unused parts as discussed in the Bill of Materials section (section 3.4). Additionally, minimising the PCB size can significantly lower manufacturing costs. This can be achieved by strategically placing components to maximise board utilization, rather than placing them arbitrarily. Finally, designing for single-sided assembly wherever possible helps reduce assembly complexity and cost, making the manufacturing process more efficient.

6.4 Dealing With Mismatching Motor Parameters

When two motors have different parameters, they will not respond identically to the same input signals. This mismatch can cause the Micromouse to deviate off course instead of moving in a straight line. To mitigate this issue, a control system can be implemented that incorporates a feedback loop that includes a controller and sensors such as encoders or current sensors. The feedback loop enables real-time monitoring and adjustment of each motor, allowing the system to autonomously compensate for any discrepancies and maintain straight-line motion.

6.5 Motor Fault Handling: Short Circuits

If the motor terminals are short circuited, this will cause a large current to flow through the terminals and through the IC of the motor driver because there is no resistance across the terminals. This excessively high current can lead to overheating or even extensive damage to the IC. However, motor driver ICs such as the DRV8833PWPR which have built-in protective circuitry such as overcurrent protection and thermal shutdown. These featured safeguard the IC and potentially the rest of the power circuitry from damage in the event of a short circuit across the motor terminals.

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